I · WHY · RESEARCHERS AND LABS · 0.2

FarmNotary

A notary for published simulation runs.
The experiment stays off-chain. The record does not.

github.com/Dooders/FarmNotary

the record, not the result

I · WHY · AUDIENCE

If someone else has to check the run, this is for you.

THIS IS FOR YOU

- You publish simulation or computational-experiment artifacts.
- A reviewer should not have to trust a laptop folder, a zip, or a Drive link.
- You want language for what the bytes allow — and a hard stop after that.

NOT THE PITCH

- This is not “put the simulation on a chain.”
- Not a score, a rank, or a lab reputation token.
- A notarized run is not a scientifically correct run.

I · WHY · THE GAP

The official copy still lives on someone's laptop.

FETCH

Get the official artifacts without asking the author for a folder.

REHASH

Check those bytes against a record that is not the zip itself.

DATE

They cannot see when that exact file was published, except by trusting the email.

Narrower than "the science is right." Broader than `sha256sum`.

A hash says the files match. A notary says what that match may mean.

A HASH TOOL

These files match this digest.

That is the entire claim.

A RESEARCH NOTARY

- Which files were admitted, by a recorded allowlist.
- How many were left out — without printing their names.
- Whether the design was stamped before the artifacts existed.
- Whether those exact bytes existed by time T.
- What you may claim. What you may not.

I · WHY · WHAT IT IS

A dated official record. You do not run a chain.

01 · WRITE

A canonical `manifest.json`: code identity, config, and artifact hashes.

02 · PIN

Optional remote pin, so the CID cites a tree instead of a laptop.

03 · STAMP

OpenTimestamps calendars batch the content hash into Bitcoin. No contract, no keys, no gas.

Immutability is not correctness. A manifest can perfectly notarize a wrong result.

Build what a stranger can check.

IN

- Canonical run and campaign manifests
- Publish profiles and a privacy denylist
- Optional remote pin through an IPFS pinning service
- A keyless Bitcoin timestamp via OpenTimestamps
- Claim card, paper-pack appendix, static index
- GitHub Action and a small Python API

OUT

- not claimed: scientific correctness
- Stopping the file drawer
- Publishing ballots or individual choices
- Scores, rankings, or a protocol identity
- Running calendars, contracts, or pin services
- Running the simulation in the EVM

II · RECORD · OFFICIAL VS PRIVATE

Nothing is hashed until you say what may leave.

STAYS PRIVATE

Any path containing ballot, vote, voter, individual_choice, or private stays out — even if a glob would have admitted it.

OFFICIAL RECORD

Config, code identity, aggregate metrics, winner allocations. Named profiles so labs do not invent the allowlist from scratch.

01 PROFILE

```
farm-notary manifest --run-dir path/to/run --profile consensus
```

Unmatched files raise a count. Their names are never printed.

II · RECORD · SOLO PATH

Three commands. Then a claim card.

01 HASH

```
farm-notary manifest --run-dir path/to/run --profile consensus --config config.json
```

02 VERIFY

```
farm-notary verify --run-dir path/to/run
```

03 STAMP

```
farm-notary anchor --run-dir path/to/run --backend ots --pin-remote pinata
```

Hours later, `farm-notary upgrade --run-dir path/to/run` finishes the Bitcoin attestation. A pending calendar is not L0.

Reviewers read the card, not the exit code.

FARM-NOTARY VERIFY | CLAIM CARD

```
level: none — no earned ladder level
next:  L0 — these bytes existed by time T; Bitcoin headers not verified by this tool (missing: Bitcoin attestation)
•  tamper-evident record          — pass
•  existed by time T              — pending
•  pre-specified design           — missing
•  bitwise reproducible (scoped)  — 6/6, ignored: *.mp4; byte-identical on x86-64 Linux in a pinned environment
•  not claimed: scientific correctness
```

II · RECORD · A STRANGER

Missing is not failure.

01 FETCH

```
ipfs get <cid> # or obtain the run directory another way
```

02 READ

```
farm-notary verify --run-dir <dir>
```

03 OPT IN

```
farm-notary verify --run-dir <dir> --verify-derived
```

Exit 0 means no attempted check failed — not that every claim was earned, and not that the science is right. Use step 03 only if you trust the recorded commands.

Earn a rung. Do not print one you did not earn.

NONE	No Bitcoin attestation yet, or the record does not rehash.	Not a failed experiment.
L0	These bytes existed by time T. Proof commits to the content hash and carries a Bitcoin-height attestation.	Not header verification. Use `ots verify`.
L1	Re-execution is specified: command, git SHA, environment fingerprint.	Not a completed re-run.
L2	Seed not grindable after the plan is anchored.	Not scientific correctness. Needs a live or fixture beacon compare.
L3	A Sigstore keyless signature is on the reproduction receipt. Identity is recorded, not proven independent.	Not independently reproduced.

III · EVIDENCE · PRE-SPECIFIED DESIGN

Stamp the plan before the artifacts exist.

01 PLAN

```
farm-notary precommit --config config.json --command "python run.py --out {run_dir}" --backend ots
```

02 BIND

```
farm-notary manifest --run-dir path/to/run --profile consensus --precommit precommit.json
```

A timestamp taken after the result is known is not pre-registration. FarmNotary will still stamp the favorable run. It will not call that stamp a pre-specified design.

Re-run the seed. Keep the claim inside the machine that matched.

DEMONSTRATED

8/8

100 trials, 300 voters, 8 candidates, seed 0. Summary statistics recompute from `trials.csv`.

The only sentence the tool may emit: *byte-identical on x86-64 Linux in a pinned environment*.

NOT A CLAIM

- Cross-hardware bitwise identity.
- Linux ARM smoke cell: 6/10. macOS ARM: 2/10.
- A classified byte-diff — path, timestamp, float print, video encoder — is **not a science failure**.

A multi-seed figure needs a parent record, not a folder of zips.

01 SWEEP

```
farm-notary campaign --name consensus-sweep --run-dir runs/seed-0 --run-dir runs/seed-1 --out sweep/
```

02 APPENDIX

```
farm-notary paper-pack --campaign sweep/ --out sweep/appendix.md
```

03 INDEX

```
farm-notary index --registry docs/registry.md --campaign sweep/
```

CID	content-addressed run
CONTENT HASH	canonical body; stamps excluded
BITCOIN ATTESTATION	block N, or pending
ALLOWLIST / UNMATCHED	the policy is part of the claim
READER LADDER	— do not cite Ln from the PDF

IV · START · COST AND PUBLISH

Free to stamp. Impossible to unpublish.

MACHINE

Python ≥ 3.9. Core is stdlib-only: manifest, verify, IPFS client, dry-run.

LIVE STAMP

The `[ots]` extra. Public calendars. No key, no gas, Bitcoin-backed. EAS stays experimental.

DEFAULT

CLI `anchor` and `precommit` are dry-run. Nothing is hashed until an allowlist is declared. A remote pin cannot be pulled back.

IV · START · LAB PATH

Add the Action. Fail the job if the card fails.

DOODERS/FARMNOTARY@V1.0.0 PHASE: NOTARIZE

```
- uses: dooders/FarmNotary@v1.0.0
  with:
    phase: notarize
    run-dir: results
    profile: consensus
    pin-remote: pinata
    backend: ots
```

Recommended: `phase: precommit` at job start, `phase: notarize` after the run. Pin `@v1.0.0` or a commit SHA.

IV · START · WHAT WE WILL NOT CLAIM

An anchor proves existence, not correctness.

REFUSED

- No “verified result” badge. A printed Ln is not a finding.
- No scoreboard, trusted-lab registry, or protocol token.
- No inventory of unpublished runs.

STILL TRUE

- Cherry-picking is out of scope. Mislabeling a post-hoc stamp is not.
- The author’s own reproduction is bitwise (scoped), not independently reproduced.
- Cross-hardware bitwise identity is not a claim.

IV · START

Install the 0.2 line. Hash one run this afternoon.

01	INSTALL	`pip install "farm-notary[ots]"`
02	HASH	`farm-notary manifest --run-dir path/to/run --profile consensus`
03	READ	`farm-notary verify --run-dir path/to/run`

github.com/Dooders/FarmNotary

Docs: PRINCIPLES · CLAIMS · DESIGN. Apache-2.0. Current release: 1.0.0 on PyPI.